

Aisha Abdur Rahim

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TECHNICAL SKILLS

Backend Engineering: Python, Node.js, Express, FastAPI, Flask, Django, RESTful APIs, GraphQL
Frontend Development: React.js, Next.js, TypeScript, JavaScript (ES6+), Redux, Supabase, HTML/CSS
Databases & DevOps: PostgreSQL, SQL, MongoDB, Firebase, Git, GitHub, AWS, GCP, Docker, CI/CD
Core Architectural Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, System Design

EXPERIENCE

- Northeastern University** San Jose, CA
Research Assistant (HCAI Lab) Jan 2026 – Apr 2026
- Architected and deployed the full backend infrastructure for AEGIS, a 5-agent adaptive system utilizing **FastAPI and Python**, executing deterministic microservice routing logic that reduced communication errors by **15%**.
 - Engineered an automated synthetic evaluation data pipeline across **15 scripts** to benchmark structural behavioral metrics, achieving a reliable rule-following precision system rate of **0.91**.
- Graduate Teaching Assistant (Algorithms)* Sep 2025 – Dec 2025
- Mentored **50+ graduate students** in algorithmic runtime complexity optimization, covering graph traversals (BFS/DFS), dynamic programming, sliding windows, and memory-efficient data structures.
 - Conducted **10+ mock technical interviews** evaluated against optimal system performance, problem-solving patterns, and runtime analysis.
- Sky IT Ltd** Dubai, UAE
Software Engineer Apr 2023 – Sep 2023
- Engineered robust mobile application layer workflows in Kotlin, orchestrating smooth data hydration by seamlessly integrating **5+ RESTful APIs** with downstream backend services.
 - Triaged and hot-fixed a critical thread-blocking architectural bug within the core runtime layer, successfully reducing production system crash rates by **25%** and driving up session performance metrics.
 - Collaborated cross-functionally across distributed teams of PMs, designers, and backend microservice engineers within an Agile framework to manage production distribution pipeline lifecycles.

PROJECTS

- MyReccBox** | *Next.js, TypeScript, Supabase, PostgreSQL, Vercel, REST APIs* | [Live](#) May 2026 – Present
- Designed and implemented a full-stack recommendation aggregator within a rigorous **3.5-hour hackathon constraint**, scaling deployment via Vercel for instant centralized access.
 - Structured an optimized relational PostgreSQL schema in Supabase to orchestrate multi-user data isolation, leveraging web-tier Supabase subscriptions to deliver low-latency, real-time UI data synchronization.
- PronunciAid** | *JavaScript, Chrome Extension API, HTML/CSS* | [Chrome Web Store](#) Dec 2025
- Shipped an interactive browser utility utilizing context menus and content scripts, refining request orchestration architectures to deliver a **3× improvement** in system response speed (15s to 2s).

EDUCATION

- Northeastern University** San Jose, CA
Master of Science in Computer Science (GPA: 4.0) Sep 2024 – May 2026
- Middlesex University Dubai** Dubai, UAE
Bachelor of Science (Hons) in Information Technology (First Class Honors) Sep 2018 – Jun 2021

PUBLICATIONS

- AI-Generated Meta-Feedback in Student Learning and Collaboration* | PETRA 2026 (In print) May 2026
- Battling the Rise of Procrastination with the Implementation of Agile Methodology* | ICMECE 2022 Nov 2022